

ABSTRACT

THE APPLICATION OF HIDDEN MOMENTUM AND GAUGE THEORY TO MAGNETIC MONOPOLE-LIKE MODELS

We demonstrate an application of the concept of hidden momentum and the mathematical techniques of gauge theory to three magnetic monopole-like models—the Dirac monopole, the Wu-Yang fiber bundle, and the Banderet potential—and show that this hidden momentum cancels out their respective field momentums, which resolves an apparent violation of the center-of-energy theorem.

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THE APPLICATION OF HIDDEN MOMENTUM AND GAUGE THEORY
TO MAGNETIC MONOPOLE-LIKE MODELS

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INTRODUCTION

It is well-known that in electromagnetism both momentum and angular momentum are carried in the electric and magnetic fields. Using the notation and units of [1] the field momentum and field angular momentum are given by

$$\mathbf{P}_{\text{EM}} = \frac{1}{4\pi} \int (\mathbf{E} \times \mathbf{B}) d^3x , \quad (1)$$

and

$$\mathbf{L}_{\text{EM}} = \frac{1}{4\pi} \int \mathbf{x} \times (\mathbf{E} \times \mathbf{B}) d^3x . \quad (2)$$

We have set the speed of light to unity ($c = 1$) throughout the paper, otherwise the pre-factor for both equations would be $\frac{1}{4\pi c}$.

In reference [2] Shockley and James considered a magnetic dipole composed of two counter-rotating electrically charged disks and two opposite electric charges. As the disks are brought to rest, there seems, at first, to be a violation of momentum conservation. To account for this discrepancy of momentum conservation, Shockley and James proposed that there must be some “hidden” momentum in the rotating disks to preserve the conservation law. Subsequently, Coleman and Van Vleck [3] demonstrated that there was a previously unaccounted for momentum in the system addressing the paradox raised in [2]. The problem revealed itself to be a discrepancy arising from a mistaken employment of Newtonian mechanics instead of relativistic mechanics. As they state, “either force is the time rate of change of (canonical) momentum, or force is mass times the acceleration of the center

of energy. In Newtonian mechanics, these two kinds of force are trivially equivalent. In relativistic mechanics...they are different” (Coleman and Van Vleck [3]). They first noted that the changing magnetic field induces an electric field which exerts a force on the particle $\mathbf{F} = e(\frac{\mathbf{r}}{r^3} \times \frac{d\mathbf{M}}{dt})$, which seems, at least initially, to be unbalanced by an equal and opposite back-force that acts on the magnet (from the particle). Here, e refers to the electric charge, c the speed of light, $\mathbf{r}$ the position vector and $\mathbf{M}$ is the magnetization of the magnet. It is exactly the term $\sum \frac{1}{2}v_a\mathbf{r}_a$, with v_a and $\mathbf{r}_a$ being the velocity and position of a particle “a”, which, when applying the conservation of momentum equation $d\mathbf{p}/dt = 0$, this term generates an additional force term which cancels the force that appears in the original setup. Coleman and Van Vleck did the following calculation:

$$M \frac{d\mathbf{X}_m}{dt} = \mathbf{P}_m - \sum e_a e r^{-3} (\mathbf{r} \cdot \mathbf{v}_a) \mathbf{r}_a \quad (3)$$

and where $\mathbf{X}_m$ is the position of the center of energy of the magnet, with the subscript “m” referring to the magnet. Then rewriting this as

$$M \frac{d\mathbf{X}_m}{dt} = \mathbf{P}_m - e r^{-3} \mathbf{r} \times \mathbf{M} \quad (4)$$

and applying the conservation of total momentum and taking the derivative with respect to time on both sides

$$M \frac{d^2\mathbf{X}_m}{dt^2} = -e r^{-3} \mathbf{r} \times \frac{d\mathbf{M}}{dt} \quad (5)$$

which ensures that the system is balanced/the net force is zero. It is exactly that when we treat the system in terms of relativistic mechanics which generates this additional force, and which leads to the concept of hidden momentum—we find that treating our system as occurring in relativistic spacetime leads to an additional momentum in the particles in a system which cancels the momentum stored in the fields.

In contemporary physics, hidden momentum is a well-known concept that has been discussed in various articles (see the resource letter [4] for an extensive list) and textbooks [5]. To see an especially concise introduction to the concept, see [5]—here, the introduction of relativistic Lorentz factors “ γ ” into the calculation of momentum for charges moving in a rectangular loop of wire introduces an additional momentum term, the titular hidden momentum. We obtain the traditional equation for hidden momentum through the following reasoning:

$$p_{rel} = \gamma_t m N_t v_t - \gamma_b m N_b v_b = \frac{m I l}{q} (\gamma_t - \gamma_b) \quad (6)$$

where t refers to the particles in the top segment and b the lower segment and “ N ” referring to the number of charges, “ v ” the velocity, “ l ” the length of the rectangle, and “ I ” the current. Then,

$$\gamma_t m - \gamma_b m = q E w \quad (7)$$

which holds as the particles in the top segment move faster and the particles going up the left side gain energy equal to the work done by the electric force, then substituting the former into the latter we have

$$\boldsymbol{p}_{hid} = \boldsymbol{m} \times \boldsymbol{E} \quad (8)$$

where $\boldsymbol{m}$ is the dipole moment of the loop. [5] In this paper we show that hidden mechanical momentum has a role in the conservation of momentum and angular momentum in three magnetic monopole-like models.

MONOPOLE MODELS

The magnetic monopole-like models considered in this paper are entirely within the original regime of classical electrodynamics, they do not include free magnetic monopole sources or involve doubling of the potentials, but rather are in the context of the standard single scalar and single vector potential

$$\mathbf{E} = -\nabla\phi_e - \frac{\partial\mathbf{A}}{\partial t} \quad ; \quad \mathbf{B} = \nabla \times \mathbf{A} \quad , \quad (9)$$

so that one has $\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0$. In the following three subsections we will discuss how Dirac and others dealt with this zero divergence of the magnetic field in the case where the potentials are given by (9). In all three cases, the zero divergence of $\mathbf{B}$ is achieved by balancing the outward flux of the magnetic Coulomb field with some compensating inward flux.

Dirac String Potential

The oldest and most well-known model for a monopole-like field within the context of standard electrodynamics is due to Dirac [6, 7] (several good reviews can be found in [8, 9, 10, 11, 12, 13]); it is based on the string potential:

$$\mathbf{A}_{\pm}(\mathbf{x}) = \frac{g}{r} \left(\frac{\pm 1 - \cos \theta}{\sin \theta} \right) \hat{\varphi} \quad . \quad (10)$$

Equation (10) shows two possible setups of the Dirac vector potential, $\mathbf{A}_+(\mathbf{x})$ and $\mathbf{A}_-(\mathbf{x})$. The vector potentials have a linear string singularity: $\mathbf{A}_+(\mathbf{x})$ is singular along the $-z$ axis (*i.e.* $\theta = \pi$) but is regular along the $+z$ axis, and

$\mathbf{A}_-(\mathbf{x})$ is singular all along the $+z$ axis (*i.e.* $\theta = 0$), but regular along the $-z$ axis. Since electromagnetism refers to a $U(1)$ Abelian gauge group, the vector potentials in (10) are related by the gauge transformation $\mathbf{A}_+(\mathbf{x}) = \mathbf{A}_-(\mathbf{x}) + \nabla\alpha$ with the non-single-valued gauge function $\alpha = 2g\varphi$, which yields $\nabla\alpha = (2g)/(r \sin\theta)\hat{\varphi}$. Further details on this gauge transformation are included at the end of this section. Except on the z -axis, the curl of the vector potentials in (10), gives a Coulomb magnetic field $\mathbf{B}^{\text{Coulomb}} = \nabla \times \mathbf{A}_{\pm}(\mathbf{x}) = g \frac{\mathbf{r}}{r^2}$.

If we take a closer look at the string singularities in $\mathbf{A}_+(\mathbf{x})$ and $\mathbf{A}_-(\mathbf{x})$, it can be shown that each potential has a physical interpretation of an infinitesimally thin solenoid which carries an inward magnetic flux of $-4\pi g$. This inward flux balances the outward magnetic flux of $4\pi g$ coming from the Coulomb magnetic field. A monopole potential can also be created by averaging the two potentials in (10) as $\frac{1}{2}(\mathbf{A}_+ + \mathbf{A}_-) = \frac{g}{r} \frac{\cos\theta}{\sin\theta} \hat{\varphi}$. After this, we see that this configuration has string singularities along both the positive and negative z axes, with an inward flux of $-2\pi g$ along each string singularity for a total of $-4\pi g$. To formulate this interpretation of (10) as an infinitesimally thin solenoid, we rewrite the vector potentials in cylindrical coordinates. This turns the expression in (10) into

$$\mathbf{A}_{\pm}(\mathbf{x}) = \frac{g}{\rho} \left(\pm 1 - \frac{z}{\sqrt{\rho^2 + z^2}} \right) \hat{\varphi} . \quad (11)$$

Next, regularize the string singularities of (11) as ¹

$$\mathbf{A}_{\pm}^{\varepsilon}(\mathbf{x}) = \frac{g\Theta(\rho - \varepsilon)}{\rho} \left(\pm 1 - \frac{z}{\sqrt{\rho^2 + z^2 + \varepsilon^2}} \right) \hat{\varphi}, \quad (12)$$

where $\Theta(x)$ is a step function (*i.e.* $\Theta(x) = 0$ if $x < 0$ and $\Theta(x) = 1$ if $x \geq 0$) and ε is a small quantity which can be taken to zero. The regularized vector potential in (12) removes the string singularity. Taking the curl of $\mathbf{A}_{\pm}^{\varepsilon}(\mathbf{x})$ and letting $\varepsilon \rightarrow 0$ gives the magnetic field

$$\begin{aligned} \mathbf{B} &= \lim_{\varepsilon \rightarrow 0} (\nabla \times \mathbf{A}_{\pm}^{\varepsilon}) = g \frac{\rho \hat{\rho} + z \hat{\mathbf{z}}}{(\rho^2 + z^2)^{3/2}} \pm 2g \frac{\delta(\rho)}{\rho} \Theta(\mp z) \hat{\mathbf{z}} \\ &= g \frac{\hat{\mathbf{r}}}{r^2} \pm 4\pi g \delta(x) \delta(y) \Theta(\mp z) \hat{\mathbf{z}}. \end{aligned} \quad (13)$$

In the last step, we have converted back from cylindrical coordinates to spherical coordinates for the Coulomb term and to Cartesian coordinates for the delta function term.

The Coulomb term in (13) gives an outward magnetic flux of $4\pi g$, while the delta function term in (13) gives an inward magnetic flux of $-4\pi g$. It follows that the magnetic field in (13) contains zero net flux since the outward Coulomb flux is canceled by the inward solenoidal flux. Appendix A contains the details of this calculation.

The result for the magnetic field in (13) can be found in several review articles and monographs [8, 9, 10, 11, 12, 13, 14]. To convert the

¹The details of the calculation starting with (12) and leading to the final result in (13) are found in Appendix D of reference [8].

magnetic field in (13) to a Coulomb magnetic field, the solenoidal part of the magnetic field is "hidden" from any electric charge, q , that is placed near the magnetic charge, g . Hiding the string part of the magnetic field in (13) is accomplished by applying the Dirac quantization condition [6, 13]

$$qg = n \frac{\hbar}{2}, \quad (14)$$

i.e. the product of the charges q and g is equal to an integer (*i.e.* n) multiple of Planck's reduced constant divided by 2. There are multiple procedures to obtain the quantization condition in (14) which can be found in various textbooks, review articles, and monographs [1, 8, 9, 10, 11, 12, 13, 14]. For example, (14) can be derived by requiring that the Aharonov-Bohm phase [15] be undetectable *i.e.* that

$$\begin{aligned} \exp \left[- \left(\frac{iq}{\hbar} \right) \oint \mathbf{A}_{string} \cdot d\mathbf{l} \right] &= \exp \left[- \left(\frac{iq}{\hbar} \right) \int_S \mathbf{B}_{string} \cdot d\mathbf{a} \right] \\ &= \exp \left[\left(\frac{iq}{\hbar} \right) 4\pi g \right] = 1 \rightarrow \frac{4\pi qg}{\hbar} = 2\pi n \end{aligned} \quad (15)$$

which leads to (14).

Next, we calculate the current density associated with the magnetic field in (13) via $\mathbf{J}_e = \frac{\nabla \times \mathbf{B}}{4\pi}$. The curl of the Coulomb part of (13) results in

zero while the curl of the solenoidal string part gives

$$\begin{aligned}\mathbf{J}_e &= \nabla \times (\pm g \delta(x) \delta(y) \Theta(\mp z) \hat{\mathbf{z}}) \\ &= \pm g \Theta(\mp z) [\delta(x) \delta'(y) \hat{\mathbf{x}} - \delta(y) \delta'(x) \hat{\mathbf{y}}] .\end{aligned}\tag{16}$$

This result will be used later when we calculate the hidden momentum which is the negative of the volume integral of the product of the scalar potential of the electric charge q , and the current density $\mathbf{J}_e$ associated with the Dirac string *i.e.* $\mathbf{P}_{hid} = - \int \phi_e \mathbf{J}_e d^3x$.

Finally, it should be noted that the $\mathbf{B}$ field in (13) can be physically interpreted as a line of magnetic dipoles that extends from the origin to infinity. One can write $\mathbf{B}$ in terms of an $\mathbf{H}$ field and a magnetization term, $\mathbf{M}$, as

$$\mathbf{B} = \mathbf{H} + 4\pi \mathbf{M} .\tag{17}$$

The $\mathbf{H}$ field can be written as the gradient of a scalar potential $\mathbf{H} = -\nabla \phi_m$ (see [1]) and choosing a Coulomb scalar potential, $\phi_m = \frac{g}{r}$ gives a Coulomb $\mathbf{H}$ -field $\mathbf{H} = g \frac{\hat{\mathbf{r}}}{r^2}$. The magnetization vector provides the delta function term in (13) with $\mathbf{M} = \pm g \delta(x) \delta(y) \Theta(\mp z) \hat{\mathbf{z}}$. The curl of $\mathbf{M}$ gives the current density in (16) $\mathbf{J}_e = \nabla \times \mathbf{M} = \pm g \Theta(\mp z) [\delta(x) \delta'(y) \hat{\mathbf{x}} - \delta(y) \delta'(x) \hat{\mathbf{y}}]$.

It should be noted that there is an explicit geometric and topological construction of the Dirac monopole [16]: it is an example of a principal G fiber bundle. Figure 1 demonstrates the necessary mappings involved in defining a fiber bundle.

$$\begin{array}{ccc}
p^{-1}(U_x) & \xrightarrow[\cong]{\Psi_{U_x}} & U_x \times F \\
p \downarrow & & \downarrow \text{proj} \\
U_x & = & U_x
\end{array}$$

Figure 1. The commutative diagram for a fiber bundle [17]. Note that p is the inverse of p^{-1} and $proj$ is the projection map that takes the product $U_x \times F$ to the first component U_x . As a commutative diagram, whether you start from $p^{-1}(U_x)$ or $U_x \times F$ the map ends at U_x .

First, note that “ B be [a] connected space with a basepoint $b_0 \in B$, and $p : E \rightarrow B$ be a continuous map.” A fiber bundle is defined as thus: “The map $p : E \rightarrow B$ is...a fiber bundle, with fiber F if it satisfies the following properties: (1) $p^{-1}(b_0) = F$ (2) $p : E \rightarrow B$ is surjective (3) For every point $x \in B$ there is an open neighborhood $U_x \subset B$ and a ‘fiber preserving homeomorphism’ $\Psi_{U_x} : p^{-1}(U_x) \rightarrow U_x \times F$, that is a homeomorphism making the diagram above commute” [17]. The distinguishing feature of a fiber bundle is its way to describe both local and global data, most clearly seen in the contrast between a cylinder, seen in Figure 2, and a Mobius strip, seen in Figure 3. Both can be defined as a circle bundle, but the Mobius strip has twisted fibers attached to it—locally, we see just the circle in each local neighborhood at the base of each fiber, but globally we detect the fibers’ twisting. Finally, we note that the set of all $\{(U_x, \Psi_{U_x})\}$ is a local

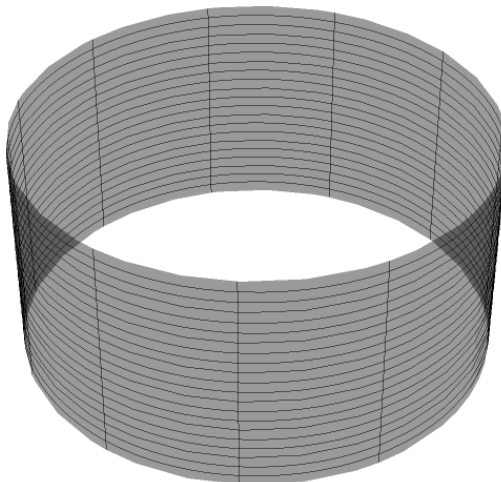


Figure 2. This is the cylinder, which is a trivial fiber bundle created from the circle bundle by attaching lines (fibers) around a circle without any twisting

trivialization of the bundle. For our purposes it is useful to also use transition functions, defined as a mapping between two coordinate charts. Formally, given coordinate charts $\{U_\alpha, \Psi_{U_\alpha}\}$ and $\{U_\beta, \Psi_{U_\beta}\}$ for a manifold M such that their intersection $U_\alpha \cap U_\beta$ is non-empty, we define the transition function as $\sigma_{\alpha,\beta} : \Psi_{U_\alpha}(U_\alpha \cap U_\beta) \rightarrow \Psi_{U_\beta}(U_\alpha \cap U_\beta)$ as $\sigma_{\alpha,\beta} = \Psi_{U_\beta} \circ \Psi_{U_\alpha}^{-1}$. Transition functions act as the “glue” between local trivializations (we glue the fibers together) that gives the global structure of the fiber bundle and we can view these functions as gauge transformations in a local sense. In the Dirac monopole, we are specifically dealing with a principal fiber bundle, which has additional structure, where G denotes a topological group, in our

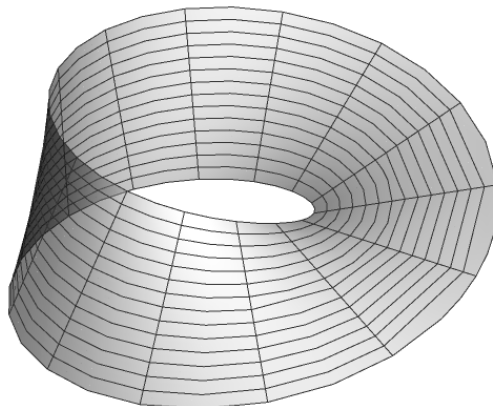


Figure 3. The Möbius strip, visualized. We can see that if we take a circle around the equator of the Möbius strip and then extend lines outward in a twisting fashion around until they meet again we achieve the Möbius strip, which is a non-trivial fiber bundle due to this twisting. Locally, the vertical strips look identical to the Möbius strip (imagine just taking one of these vertical slices from each and laying them on a flat table), but globally we see the twisting nature

case $U(1)$, with a continuous right action $P \times G \rightarrow P$ such that G preserves the fibers of P and acts freely and transitively on them in such a way that for each $x \in X$ and $y \in P_x$, the map $G \rightarrow P_x$ sending g to yg is a homeomorphism. In the Dirac model, we see that we must consider two regions, one $U = \mathbb{R}^3 - \{\text{negative } z\text{-axis}\}$ and one $V = \mathbb{R}^3 - \{\text{positive } z\text{-axis}\}$. It is apparent that the intersection of U and V results in a disagreement between their magnetic vector potentials. First off, suppose we have a basis $\mathbf{e}_v$, we then define a gauge transformation as follows: using a function

$g_{uv} : U \cap V \rightarrow Gl(n; \mathbb{R})$ where we map the intersection of two maps U and V to the general linear group (group of all non-singular real $n \times n$ matrices), and thus our gauge transformation is $e_V = e_U g_{UV}$. For our specific case, we define the transformation rule for the matrix of a connection of 1-forms as $\omega' = P^{-1}\omega P + P^{-1}dP$ (here, the ω terms will be our magnetic vector potential terms and the P terms are non-singular $n \times n$ matrix functions). Since we are in the $U(1)$ gauge, ω is abelian, and so we simply subtract the first term with the second to determine $P^{-1}dP$. We find that ultimately, to remediate this, we use the electromagnetic bundle $g_{UV}(x) = \exp\left(\frac{ie}{\hbar} f_{uv}\right)$ with connection $\omega_j := -(\frac{ie}{\hbar} A_j)$ [16]. Explicitly doing the calculation, we can construct a transition map/function for this monopole bundle $g_{uv} = \exp(-\frac{2ieq\phi}{\hbar})$ by the following procedure:

$$\mathbf{A}_u - \mathbf{A}_q = q(1 - \cos(\theta))\mathbf{d}\phi + q(1 + \cos(\theta))\mathbf{d}\phi = 2q\mathbf{d}\phi = \mathbf{d}\beta \quad (18)$$

This is a differential one form, denoted so by the bolded $\mathbf{d}$. A differential form “is a (continuous) tensor field whose value at each point [on our manifold] is an alternating tensor.” [18] We note that $\beta = 2q\phi$ and thus we can construct a gauge transformation function (which is known as a transition function to mathematicians)

$$g_{UV} = \exp\left(-\frac{ie\beta}{\hbar}\right) = \exp\left(-\frac{2ieq\phi}{\hbar}\right) \quad (19)$$

We note that this equation must satisfy the Dirac quantization condition in order to be single-valued and therefore a well-defined fiber bundle that describes the Dirac monopole model system. Furthermore, it is shown in [16] that

$$\frac{i}{2\pi} \int_{M^2} \theta^2 = integer \quad (20)$$

when the complex line bundle is the tangent bundle to oriented closed surface M^2 (θ is our curvature). This equation is derived from a generalization of the Gauss-Bonnet theorem

$$\frac{i}{2\pi} \int \int_V \theta^2 \quad (21)$$

where we have a hermitian line bundle E , a pure, imaginary connection ω^1 and curvature θ^2 over a manifold M^n with V^2 being any closed oriented surface embedded in M^n ; the integral is an integer, as long as $H_2(M^n) \cong \mathbb{Z}$ is non-trivial (its second homology group is non-zero; for a good introduction to homology in physics see [16]) and $\frac{i\theta}{2\pi}$ is the Chern form of E ; we can think of $H_2(M^n) \cong \mathbb{Z}$ as indicating that there is a two-dimensional hole in the planar manifold, as we can “wrap around” the singularity an integer number of times (increasing the integer by one for every complete winding) which checks out with the fact that the Dirac monopole system has a singularity at the location of the magnetic monopole. It is well-known that for S^2 , $H_2(S^2) \cong \mathbb{Z}$ which indicates a topological quantization condition. Back to the case of the Dirac monopole, we achieve the same equation as (29) when our surface is S^2

(more generally, M^2). Specifically, for our monopole bundle, we have

$$\theta = - \left(\frac{ie}{\hbar} \right) (E \wedge dt + B) \quad (22)$$

which reduces in our setup as there is no electric field. The “ $\wedge$ ” term is the wedge product, defined by the following: Given $\omega \in \Sigma^k(V^*)$ and $\eta \in \Sigma^l(V^*)$ where $\Sigma^k(V^*)$ refers to a k-covector:

$$\omega \wedge \eta = \frac{(k+l)!}{k!l!} Alt(\omega \otimes \eta) \quad (23)$$

where $Alt(\omega \otimes \eta)$ is the alternating tensor product. [18]

Noting that $\mathbf{B} = (q/r^2)\partial/\partial\mathbf{r}$ we achieve the result

$$\theta = -\frac{ie}{\hbar}\mathcal{B} = -\frac{ie}{\hbar}i_{\mathbf{B}}vol^3 = -\frac{ieq}{\hbar r^2}i_{\frac{\partial}{\partial r}}vol^3 \quad (24)$$

where $\mathcal{B} = i_{\mathbf{B}}vol^3 = \mathbf{d}[q(1 - \cos\theta)d\phi]$ (vol^3 is the spherical volume form from differential geometry) and $i_{\mathbf{B}}$ refers to the interior product defined as (in local coordinates) over the vector field “ $\mathbf{B}$ ” :

$i_{\mathbf{X}}(dx_1 \wedge \dots \wedge dx_n) = \sum_{r=1}^n (-1)^{r-1} f_r dx_1 \wedge \dots \wedge \widehat{dx_r} \wedge \dots \wedge dx_n$ (where the hat denotes omission of the dx_r differential, so given vector field $\mathbf{B}$ we have

$$\mathcal{B} = i \left(\frac{q}{r^2} \frac{\partial}{\partial r} \right) r^2 \sin(\theta) dr \wedge d\theta \wedge d\phi = q \sin(\theta) d\theta \wedge d\phi = \mathbf{d}[q(1 - \cos(\theta)d\phi)] \quad (25)$$

so the partial derivative here doesn't apply to the r^2 term but rather corresponds to the dr differential that will be removed after applying the interior product.

Now, applying (20) we get

$$\left(\frac{-i}{2\pi}\right) \left(\frac{-ieq}{\hbar}\right) \int_0^{2\pi} \int_0^\pi \sin(\theta) d\theta \wedge d\phi = -\frac{i}{2\pi} \frac{ieq}{\hbar} 4\pi = \frac{2eq}{\hbar} \quad (26)$$

so any monopole bundle with a closed-oriented manifold S^2 (or M^2) and corresponding curvature yields the quantization condition, following from the general case. This, as well as the method involving the Ahronov-Bohm phase done earlier, highlight the role that topological obstructions play in monopole theories and how tools like winding number (which shows up in the Ahronov-Bohm effect), which counts the number of counterclockwise turns an object makes around an origin, and (co)homological groups, which detect these topological “holes”, reveal these topological obstructions in a manifold. Often, like for the Dirac monopole-like model, a quantization condition—whose integers directly correspond to the turns around a topological obstruction or the number of loops around a “hole” that can be made—is imposed upon a model in order to make it physically consistent. However, a reminder that the Dirac monopole is nonphysical for other reasons, as shown earlier. Cohomology is often an easier tool to work with than homology, especially when we use differential forms (see De Rham cohomology), but its uses are beyond the scope of this thesis and in this

setup homology is simple enough and more commonly used in the magnetic monopole literature. It will be demonstrated in the Wu-Yang approach section how the construction of two separate vector potentials (describing two different parts of the monopole system), with an intersection $U \cap V$ at the equator where the monopole originally lay, is a modification of the Dirac string setup in terms of fiber bundle theory, allowing us to construct magnetic potentials without any singularities in their geometry.

Banderet Potential

The Banderet formulation of magnetic charge [19] is not well-known compared to the three approaches discussed in this paper. The Banderet vector potential has the form ²

$$\mathbf{A}_{\text{band}} = -\frac{g}{r}\varphi \sin \theta \, \hat{\theta} . \quad (27)$$

Taking the curl of $\mathbf{A}_{\text{band}}$ gives [9]

$$\mathbf{B} = \nabla \times \mathbf{A}_{\text{band}} = \frac{g}{r^2}\hat{\mathbf{r}} - 2\pi g\Theta(x)\delta(y)x\frac{\hat{\mathbf{r}}}{r^2} . \quad (28)$$

The $\frac{g}{r^2}\hat{\mathbf{r}}$ term in (28) is the Coulomb magnetic field of strength g , and the second term, $-2\pi g\Theta(x)\delta(y)x\frac{\hat{\mathbf{r}}}{r^2}$, is due to the non-single-valued nature of $\mathbf{A}_{\text{band}}$. This second term carries an inward magnetic flux of $-4\pi g$. This

²There is a typo in [9] which has the vector potential (27) in the $\hat{\varphi}$ direction rather than in the correct $\hat{\theta}$ direction, which can be confirmed by direct calculation.

cancels the outward magnetic flux of $+4\pi g$ coming from the Coulomb term, $\frac{g}{r^2}\hat{\mathbf{r}}$, verifying that the net divergence of the $\mathbf{B}$ -field in (28) is zero. This calculation of the canceling of the fluxes for the Banderet potential is carried out in Appendix A. The calculations in Appendix A demonstrate that the second term in (28) is necessary to balance the ingoing and outgoing magnetic fluxes.³

The current density associated with (28) can be calculated via $\mathbf{J}_e = \frac{\nabla \times \mathbf{B}}{4\pi}$. The calculation of the current density can be found in Appendix B and gives

$$\mathbf{J}_e = \frac{g\Theta(x)}{2r^3} (x^2\delta'(y)\hat{\mathbf{z}} - xz\delta'(y)\hat{\mathbf{x}} + z\delta(y)\hat{\mathbf{y}}) . \quad (29)$$

We will use this result later to calculate the hidden momentum carried by the Banderet potential.

For pedagogical purposes, there is also an alternative approach to determining the Dirac quantization condition for the Banderet potential using the Wu-Yang fiber bundle approach: To derive the Dirac quantization condition for the Banderet potential, Shnir [20] used the Wu-Yang construction to create overlapping potentials, from which he was able to derive the Dirac quantization condition in an analogous fashion as Wu and Yang—the same method that will appear in the next section. This is a much

³We have not been able to find a more direct way to obtain this second term, nor does reference [9] give a derivation, but simply states that (28) is the appropriate $\mathbf{B}$ -field following from the Banderet vector potential (27).

more intuitive and easily visualizable way to determine the Dirac quantization condition.

Wu-Yang Fiber Bundle

The Wu-Yang approach to magnetic charge [21, 22] begins with the Dirac string potentials [6, 7] and applies the mathematics of fiber bundles to avoid the string singularities of (10). Although the end result is the same (net flux of the magnetic field is zero), and the magnetic field is almost identical to the Dirac string setup, this is an interesting geometric approach to calculating the $\mathbf{B}$ -field, using an application of Stokes's Theorem instead of directly applying the curl of the vector potential. For our setup, in simple terms, the vector potential over the northern hemisphere is defined as $\mathbf{A}_+$, while the vector potential over the southern hemisphere is defined as $\mathbf{A}_-$. There is a thin overlap region around the equator that has a width 2ϵ where $\mathbf{A}_+$ and $\mathbf{A}_-$ overlap. This is shown in Figure (4) with a cylindrical surface instead of the usual spherical surface. The vector potentials $\mathbf{A}_+$ and $\mathbf{A}_-$ will be continuously connected across the overlap region 2ϵ , for example, by a linear function. The particular form of how $\mathbf{A}_+$ and $\mathbf{A}_-$ are continuously connected is not necessary for this setup since both $\mathbf{A}_+$ and $\mathbf{A}_-$ are in the $\hat{\varphi}$ direction, and the segments of the line integral in the overlap region are in the $\hat{\mathbf{z}}$ direction. Thus, $\mathbf{A}_\pm \cdot d\mathbf{l} = 0$ in the overlap region and these segments do not contribute, independent of how $\mathbf{A}_+$ and $\mathbf{A}_-$ are connected across this region.

Using boundary conditions across the equator in Figure (4) we will demonstrate that the Wu-Yang formulation also requires a delta function magnetic field in addition to the Coulomb magnetic field. Using cylindrical coordinates, we claim that the full magnetic field is of the form

$$\mathbf{B} = g \frac{\rho \hat{\rho} + z \hat{\mathbf{z}}}{(\rho^2 + z^2)^{3/2}} - \frac{2g\delta(z)}{\rho} \hat{\rho}. \quad (30)$$

The first term is the Coulomb magnetic field, and the second term is the extra delta function magnetic field. The delta function term in (30) results in an inward magnetic flux of $-4\pi g$ to cancel the outward flux of $4\pi g$ due to the Coulomb term. The details of this calculation are found in Appendix A.

Reference [23] gives a more direct construction for deriving the result in (30), whereas here we show that the delta function term is required by boundary conditions and Stokes's theorem. We do this by evaluating Stokes's theorem for the magnetic field and vector potential, $\int \mathbf{B} \cdot d\mathbf{a} = \oint \mathbf{A} \cdot d\mathbf{l}$, for the line integral and two possible surfaces as shown in Figs. (4) (5) and (6).

The thin solid line with arrows in Figure (4) is the contour that runs from $A \rightarrow B \rightarrow C \rightarrow D = A$. The part of the contour that is the segment DA (around the back) is entirely in the northern hemisphere where the vector potential is $\mathbf{A}_+$; the segment BC (around the back) is entirely in the southern hemisphere where the vector potential is $\mathbf{A}_-$; the two short vertical segments, AB and CD cross the equator and bridge the region where $\mathbf{A}_+$ and $\mathbf{A}_-$ overlap. The two dotted lines that bracket the equator (which is the

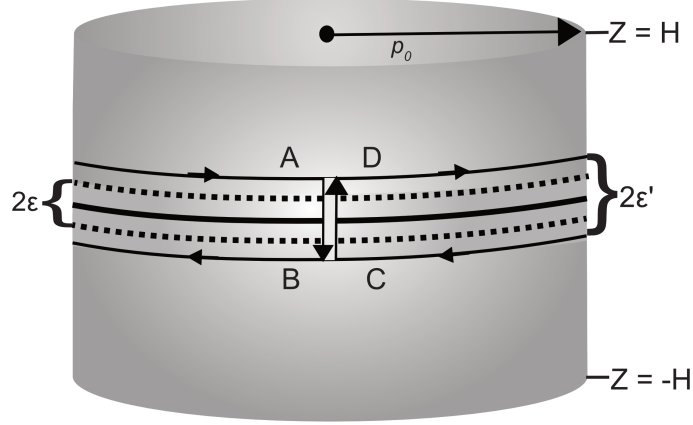


Figure 4. The line integral and cylindrical surface for evaluating $\oint \mathbf{A} \cdot d\mathbf{l}$ across the equator. This will be checked against the two possible areas in Figs. (5) and (6) for evaluating $\int \mathbf{B} \cdot d\mathbf{a}$. The thin solid line is the contour of $\oint \mathbf{A} \cdot d\mathbf{l}$, while the dotted lines denote the region of overlap between the vector potentials $\mathbf{A}_+$ and $\mathbf{A}_-$.

thicker, solid line) bound the region of width 2ϵ where the two vector potentials overlap. The solid lines of the contour above and below the equator are separated by $2\epsilon'$.

The part of the contour DA has $d\mathbf{l} = \rho_0 d\varphi \hat{\varphi}$ and the vector potential is $\mathbf{A}_+$ so we have from (11)

$$\int_D^A \mathbf{A}_+ \cdot d\mathbf{l} = \int_0^{2\pi} \frac{g}{\rho_0} \left(1 - \frac{\epsilon'}{\sqrt{\rho_0^2 + (\epsilon')^2}} \right) \hat{\varphi} \cdot \rho_0 d\varphi \hat{\varphi} = 2\pi g - \frac{2\pi g \epsilon'}{\sqrt{\rho_0^2 + (\epsilon')^2}}. \quad (31)$$

For the part of the contour AB that crosses the equator it is in the region where the vector potentials, $\mathbf{A}_+$ and $\mathbf{A}_-$, overlap, but for this contour $d\mathbf{l} \propto \hat{\mathbf{z}}$

and since both $\mathbf{A}_\pm \propto \hat{\varphi}$ and $\hat{\mathbf{z}} \cdot \hat{\varphi} = 0$ this part of the contour results zero.

The same applies to the part of the contour CD . Thus we have

$$\int_A^B \mathbf{A}_\pm \cdot d\mathbf{l} = \int_C^D \mathbf{A}_\pm \cdot d\mathbf{l} = 0. \quad (32)$$

A more careful treatment of these parts of the contour is carried out in [23] using the method laid out in [21], but the end result is the same.

Finally, the BC segment has $d\mathbf{l} = -\rho_0 d\varphi \hat{\varphi}$ since the direction of traversal is opposite to that of the segment DA , and here the vector potential is $\mathbf{A}_-$. This segment contributes to the contour as

$$\begin{aligned} \int_B^C \mathbf{A}_- \cdot d\mathbf{l} &= \int_0^{2\pi} \frac{g}{\rho_0} \left(-1 - \frac{(-\epsilon')}{\sqrt{\rho_0^2 + (-\epsilon')^2}} \right) \hat{\varphi} \cdot (-\rho_0 d\varphi \hat{\varphi}) \\ &= 2\pi g - \frac{2\pi g \epsilon'}{\sqrt{\rho_0^2 + (\epsilon')^2}} \end{aligned} \quad (33)$$

Adding all these contributions together gives

$$\begin{aligned} \oint \mathbf{A} \cdot d\mathbf{l} &= \int_D^A \mathbf{A}_+ \cdot d\mathbf{l} + \int_A^B \mathbf{A} \cdot d\mathbf{l} + \int_B^C \mathbf{A}_- \cdot d\mathbf{l} + \int_C^D \mathbf{A} \cdot d\mathbf{l} \\ &= \left(2\pi g - \frac{2\pi g \epsilon'}{\sqrt{\rho_0^2 + (\epsilon')^2}} \right) + 0 + \left(2\pi g - \frac{2\pi g \epsilon'}{\sqrt{\rho_0^2 + (\epsilon')^2}} \right) + 0 \\ &= 4\pi g - \frac{4\pi g \epsilon'}{\sqrt{\rho_0^2 + (\epsilon')^2}} \end{aligned} \quad (34)$$

Next, we evaluate $\int \mathbf{B} \cdot d\mathbf{a}$. From Figure (4) there are two possible areas spanned by the contour as shown in Figures (5) and (6). The first area

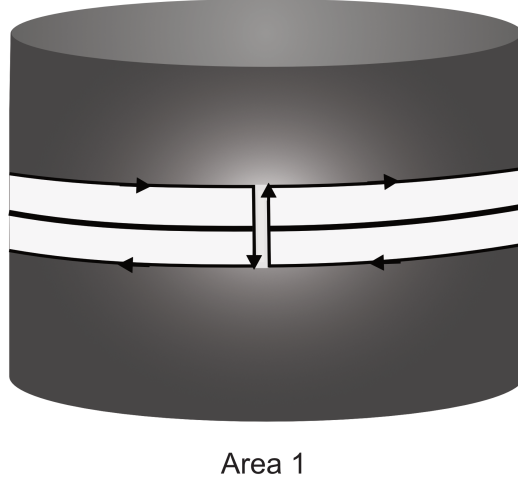
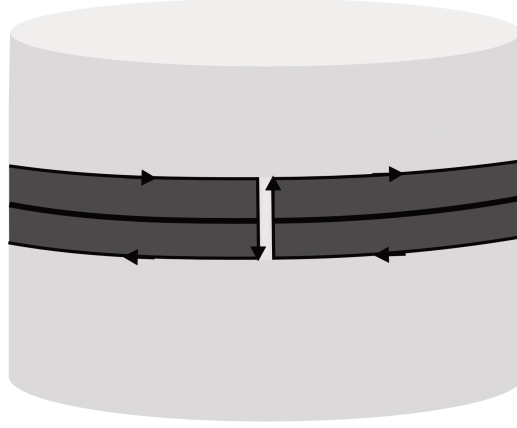


Figure 5. Area 1 is shaded in dark gray. This is one of the possible areas bounded by the line integral contour of $\oint \mathbf{A} \cdot d\mathbf{l}$ from Fig (4) for evaluating $\int \mathbf{B} \cdot d\mathbf{a}$.

is the entire surface without the strip of width $2\epsilon'$ around the equator. For this area (labeled Area 1), the additional delta function part of the magnetic field in (30) will not contribute since it lies in the thin strip that is excluded by Area 1. Thus, the magnetic flux through Area 1 is only due to the Coulomb part of (30) namely $\mathbf{B}_{\text{Coulomb}} = g \frac{\rho \hat{\rho} + z \hat{z}}{(\rho^2 + z^2)^{3/2}}$. Considering the whole cylindrical surface, this flux is the standard result

$\int_{\text{full-surface}} \mathbf{B}_{\text{Coulomb}} \cdot d\mathbf{a} = 4\pi g$. It follows that we just need to find the magnetic flux due to the Coulomb magnetic field through the thin strip and subtract this from $4\pi g$. For this thin strip of width $2\epsilon'$ the Coulomb magnetic field is $\mathbf{B}_{\text{Coulomb}} = g \frac{\rho_0 \hat{\rho} + z \hat{z}}{(\rho_0^2 + z^2)^{3/2}}$, and the infinitesimal area is



Area 2

Figure 6. Area 2 is shaded in dark gray. This is the other option for the area bounded by the line integral contour of $\oint \mathbf{A} \cdot d\mathbf{l}$ from Fig (4) for evaluating $\int \mathbf{B} \cdot d\mathbf{a}$.

$d\mathbf{a} = \rho_0 dz d\varphi \hat{\rho}$, and thus the flux through the thin strip is

$$\int_{\text{Strip}} \mathbf{B}_{\text{Coulomb}} \cdot d\mathbf{a} = g \int_0^{2\pi} \rho_0 d\varphi \int_{-\epsilon'}^{\epsilon'} dz \frac{\rho_0}{(\rho_0^2 + z^2)^{3/2}} = 4\pi g \frac{\epsilon'}{\sqrt{\rho_0^2 + (\epsilon')^2}}, \quad (35)$$

which is the missing flux due to the thin strip in Figure (4). Subtracting the result (35) from the entire cylindrical surface results in the flux through Area 1

$$\int_{\text{Area-1}} \mathbf{B}_{\text{Coulomb}} \cdot d\mathbf{a} = 4\pi g - \frac{4\pi g \epsilon'}{\sqrt{\rho_0^2 + (\epsilon')^2}} = \oint \mathbf{A} \cdot d\mathbf{l}, \quad (36)$$

which agrees with the result of the line integral, $\oint \mathbf{A} \cdot d\mathbf{l}$ in equation (34).

Next, we evaluate $\int \mathbf{B} \cdot d\mathbf{a}$ for Area 2 in Figure (6). One would expect that this would again give the same result as (36) since both areas are

bounded by the same contour. For Area 2 the area element has the opposite sign from (36), namely $d\mathbf{a} = -\rho_0 dz d\varphi \hat{\rho}$, since the direction of traversal of the contour is reversed for this area. Thus for Area 2 the Coulomb magnetic field will give the same magnitude as the result in (35) but opposite sign

$$\int_{Area-2} \mathbf{B}_{Coulomb} \cdot d\mathbf{a} = -4\pi g \frac{\epsilon'}{\sqrt{\rho_0^2 + (\epsilon')^2}} . \quad (37)$$

This result does not match the result for $\oint \mathbf{A} \cdot d\mathbf{l}$ from (34) and would appear to violate Stokes's theorem. We now show that the extra non-Coulomb piece will remedy this issue and give the value needed by Stokes's theorem.

$$\begin{aligned} \int_{Area-2} \mathbf{B}_{extra} \cdot d\mathbf{a} &= \int_{Area-2} \left(-\frac{2g\delta(z)}{\rho_0} \hat{\rho} \right) \cdot (-\rho_0 dz d\varphi \hat{\rho}) \\ &= 2g \int_{-H}^H \delta(z) dz \int_0^{2\pi} d\varphi = 4\pi g . \end{aligned} \quad (38)$$

Combining (37) and (38) gives

$$\int_{Area-2} (\mathbf{B}_{extra} + \mathbf{B}_{Coulomb}) \cdot d\mathbf{a} = 4\pi g - \frac{4\pi g \epsilon'}{\sqrt{\rho_0^2 + (\epsilon')^2}} = \oint \mathbf{A} \cdot d\mathbf{l} , \quad (39)$$

which again matches with the line integral in (34), showing that the extra, second term in (30) is needed to make Stokes's theorem work out.

Concluding this section, we calculate the current density associated with the magnetic field from (30) using $\nabla \times \mathbf{B} = 4\pi \mathbf{J}$. The Coulomb term in (30) has a curl of zero, since it is radially symmetric and naturally does not

lead to a current density. The disk term gives

$$\nabla \times \left(-\frac{2g\delta(z)}{\rho} \hat{\rho} \right) = -\frac{2g}{\rho} \frac{d(\delta(z))}{dz} \hat{\phi} = -\frac{2g}{\rho} \delta'(z) \hat{\phi} , \quad (40)$$

Thus we find a current density of

$$\mathbf{J}_e = -\frac{g}{2\pi\rho} \delta'(z) \hat{\phi} . \quad (41)$$

Since the current density is proportional to the derivative of a delta function this implies that there is a *dipole* current density at $z = 0$. The above analysis of the jump in the vector potential over a short distance and with a source that is the derivative of a delta function is essentially the magnetic version of the charge dipole layer jump of the scalar potential discussed in section 1.6 of Jackson's electrodynamics text [1].

FIELD MOMENTUM OF MAGNETIC PLUS ELECTRIC CHARGES

We begin the calculation of the field momentum by placing the field configuration of the Dirac model (13), the Banderet model (28) and the Wu-Yang model (30) of magnetic charge at the origin. Next, we place an electric charge at some point away from the origin such as $\mathbf{r}_0$. The electric field will be a Coulomb field $\mathbf{E} = q \frac{\hat{\mathbf{r}}'}{r'^2}$ with $\mathbf{r}' = \mathbf{r} - \mathbf{r}_0$ being the radial vector from the location of the electric charge. It is well known [1, 9, 8, 10, 11, 12, 14, 13] that this system of magnetic charge plus electric charge carries a field angular momentum of $\mathbf{L}_{EM} = -qg\hat{\mathbf{r}}_0$ in the Coulomb electric and Coulomb magnetic fields, with $\hat{\mathbf{r}}_0$ being a unit vector pointing from the electric charge to the magnetic charge. This result can be used to obtain the Dirac condition (14) by requiring that the magnitude of this field angular momentum is some integer multiple of the fundamental unit of angular momentum $\frac{\hbar}{2}$ *i.e.* $|\mathbf{L}_{EM}| = qg = n\frac{\hbar}{2}$ which is the Dirac condition in (14).⁴

It is also well known and easy to show that there is no linear momentum carried by the Coulomb magnetic field and the Coulomb electric field of the monopole plus electric charge system, namely

$$\mathbf{P}_{EM}^{\text{Coulomb}} = \frac{1}{4\pi} \int q \frac{\mathbf{r}'}{(r')^3} \times g \frac{\mathbf{r}}{r^3} d^3x = 0.$$

However, it will be shown in the next three subsections that the non-Coulomb parts of the magnetic fields for the

⁴Reference [24] showed that the string magnetic field gives an additional field angular momentum that voids the use of angular momentum quantization to obtain the Dirac quantization condition.

Dirac (13), Banderet (28) and Wu-Yang (30) models *do* result in a nonzero field momentum.

Field Momentum of Dirac String Plus Electric Charge

If we insert the non-Coulomb part of the Dirac string magnetic field from (13), along with the Coulomb electric field into the field momentum expression (1) we find

$$\mathbf{P}_{\text{EM}}^{\text{String}} = \frac{1}{4\pi} \int \left(q \frac{\hat{\mathbf{r}}'}{r'^2} \times (\pm 4\pi g \delta(x) \delta(y) \Theta(\mp z) \hat{\mathbf{z}}) \right) d^3x . \quad (42)$$

The specific form of the string magnetic field makes it is easiest to evaluate the integrals in (42) in Cartesian coordinates so that

$\mathbf{r}' = \mathbf{r} - \mathbf{r}_0 = (x - x_0)\hat{\mathbf{x}} + (y - y_0)\hat{\mathbf{y}} + (z - z_0)\hat{\mathbf{z}}$. The dx and dy integrations are simple to carry out using the delta functions $\delta(x)$ and $\delta(y)$. This leaves a straightforward dz integration which yields [25]

$$\mathbf{P}_{\text{EM}}^{\text{String}} = \pm gq \int_{-\infty}^{\infty} \Theta(\mp z) \frac{x_0 \hat{\mathbf{y}} - y_0 \hat{\mathbf{x}}}{(\rho_0^2 + (z - z_0)^2)^{3/2}} dz = \pm gq \frac{(r_0 \mp z_0)}{r_0 \rho_0^2} (-y_0 \hat{\mathbf{x}} + x_0 \hat{\mathbf{y}}) , \quad (43)$$

with $\rho_0^2 = x_0^2 + y_0^2$ and $r_0^2 = x_0^2 + y_0^2 + z_0^2 = \rho_0^2 + z_0^2$. This nonzero field momentum may initially seem unusual. If, for the moment, one ignores the current density in (16) then nothing in the system seems to be moving – the electric and magnetic charges are stationary, and the fields are static.

Nevertheless, there is a non-zero linear field momentum in the system. If (43) is the only momentum in the system, this would violate the

center-of-energy theorem [3, 26]. This apparent violation of the center-of-energy theorem suggests that ignoring the current density in (16) is not a feasible option. In Section IV we do not neglect the current density (16) and show that this system carries a hidden momentum in the charges and current densities, which cancels the field momentum.

Field Momentum of Banderet Potential Plus Electric Charge

Next we calculate the field momentum for the non-Coulomb part of the Banderet magnetic field – the second term in (28) – which is

$$\mathbf{P}_{\text{EM}}^{\text{Banderet}} = -\frac{2\pi qg}{4\pi} \int \Theta(x)\delta(y)x \frac{\mathbf{r}'}{(r')^3} \times \frac{\mathbf{r}}{r^3} d^3x . \quad (44)$$

We choose the electric charge to be located at $\mathbf{r}_0 = y_0\hat{\mathbf{y}}$. In attempting to calculate the most general case with an electric charge placed at a general location, we did not attain an analytic result for the integral in (44). With this choice of $\mathbf{r}_0$, we have $\mathbf{r}' = \mathbf{r} - y_0\hat{\mathbf{y}}$ and $r' = (x^2 + (y - y_0)^2 + z^2)^{1/2}$.

Using this in (44) and carrying out the dy integration gives

$$\begin{aligned} \mathbf{P}_{\text{EM}}^{\text{Banderet}} &= \frac{qgy_0}{2} \int_0^\infty dx \int_{-\infty}^\infty dz \frac{(-x^2\hat{\mathbf{z}} + xz\hat{\mathbf{x}})}{(x^2 + y_0^2 + z^2)^{3/2}(x^2 + z^2)^{3/2}} \\ &= -\frac{qgy_0}{2} \int_0^\infty dx \int_{-\infty}^\infty dz \frac{x^2\hat{\mathbf{z}}}{(x^2 + y_0^2 + z^2)^{3/2}(x^2 + z^2)^{3/2}} . \end{aligned} \quad (45)$$

In going from the first line to the second we have carried out the dz integration for the $\hat{\mathbf{x}}$ component, which gives zero since the integration range

is over all z , from $-\infty$ to $+\infty$, and the integrand, as a function of z , is an odd function. Finally we use *Mathematica* to carry out the dx and dz integrations in the last line of (45) yielding

$$\mathbf{P}_{\text{EM}}^{\text{Banderet}} = -\frac{qg\pi}{4y_0}\hat{\mathbf{z}}. \quad (46)$$

As for the Dirac string potential, we see that there is apparently an uncanceled field momentum. In Section IV we will show that this system also carries hidden momentum in the charges and currents that give rise to the full magnetic field in (28).

Field Momentum of Wu-Yang Fiber Bundle Plus Electric Charge

We now calculate the field momentum of an electric charge plus the disk magnetic field of the Wu-Yang monopole model, the second term in (30). Using cylindrical symmetry and without loss of generality, we place the electric charge q at $\mathbf{r}_0 = \rho_0\hat{\mathbf{x}} + z_0\hat{\mathbf{z}}$ giving an electric field of $\mathbf{E} = q\frac{\mathbf{r}'}{r'^3}$ where $\mathbf{r}' = \mathbf{r} - \mathbf{r}_0$. In cylindrical coordinates $r' = (\rho^2 + \rho_0^2 - 2\rho\rho_0\cos\varphi + (z - z_0)^2)^{1/2}$. With this set up the field momentum of the electric charge plus disk magnetic field is

$$\mathbf{P}_{\text{EM}}^{\text{disk}} = \frac{1}{4\pi} \int q \frac{\hat{\mathbf{r}}'}{r'^2} \times \left(-\frac{2g\delta(z)}{\rho} \hat{\rho} \right) d^3x \quad (47)$$

Using $\delta(z)$ to handle the dz integration gives

$$\begin{aligned}\mathbf{P}_{\text{EM}}^{\text{disk}} &= -\frac{qg}{2\pi} \int_0^\infty \rho d\rho \int_0^{2\pi} d\varphi \frac{[\rho\hat{\rho} - \rho_0\hat{\mathbf{x}} - z_0\hat{\mathbf{z}}]}{(\rho^2 + \rho_0^2 + z_0^2 - 2\rho\rho_0 \cos \varphi)^{3/2}} \times \frac{\hat{\rho}}{\rho} \quad (48) \\ &= -\frac{qg}{2\pi} \int_0^\infty d\rho \int_0^{2\pi} d\varphi \left(\frac{z_0 \sin \varphi \hat{\mathbf{x}} - z_0 \cos \varphi \hat{\mathbf{y}} - \rho_0 \sin \varphi \hat{\mathbf{z}}}{(\rho^2 + \rho_0^2 + z_0^2 - 2\rho\rho_0 \cos \varphi)^{3/2}} \right) .\end{aligned}$$

In going from the first line to the second we have carried out the cross product. The $\hat{\mathbf{x}}$ and $\hat{\mathbf{z}}$ components in the last line of (48) are of the same form, and carrying out the $d\varphi$ integration gives zero for both. This leaves the field momentum as

$$\mathbf{P}_{\text{EM}}^{\text{disk}} = \frac{qgz_0}{2\pi} \int_0^\infty d\rho \int_0^{2\pi} d\varphi \frac{\cos \varphi \hat{\mathbf{y}}}{(\rho^2 + \rho_0^2 + z_0^2 - 2\rho\rho_0 \cos \varphi)^{3/2}} . \quad (49)$$

Using *Mathematica* the integration of the $\hat{\mathbf{y}}$ component gives

$$\mathbf{P}_{\text{EM}}^{\text{disk}} = -\frac{qgz_0}{\rho_0 r_0} \hat{\mathbf{y}} \quad (50)$$

$\mathbf{P}_{\text{EM}}^{\text{disk}} \neq 0$ except for two special cases when $z_0 = 0$ for q in the z -plane, and $\rho_0 = 0$ for q along the z -axis. In both cases $\mathbf{P}_{\text{EM}}^{\text{disk}} = 0$. For the case $\rho_0 = 0$, the φ integral in (49) is zero, and the ρ integral is finite.

HIDDEN MECHANICAL MOMENTUM

In the above sections IIIA, IIIB, IIIC we found that the Dirac, Banderet, and Wu-Yang models all have a non-zero field momentum in the electric and magnetic fields when an electric charge, q , is placed in the vicinity of the magnetic fields in equations (13), (28), and (30). Initially, this would seem to violate the center-of-energy theorem [3, 26] since this system has a net momentum despite static fields and charges. [2] was the first to consider such a case which led to apparent paradoxes/problems, which were resolved [3] via “hidden” mechanical momentum *i.e.* momentum carried by the currents and charges that make up the system. Below we show that each of these three monopole models – Dirac, Banderet, and Wu-Yang – carries such a hidden momentum, which cancels the field momentum and resolves the issue concerning the center-of-energy theorem.

Good discussions of “hidden” mechanical momentum can be found in references [27, 28]. An illuminating example of “hidden” momentum is given in [5] and [29] which involves a rectangular current loop of moving charges embedded in a uniform electric field. In this rectangle loop example, the hidden momentum comes from the γ factor in the relativistic expression for momentum. The γ factor indicates that hidden momentum is an effect that arises from special relativity. One can generalize the specific example given in [5] and [29] to give an expression for the “hidden” mechanical momentum

stored in a system of steady currents and static electric field:

$$\mathbf{P}_{\text{hid}} = - \int \phi_e \mathbf{J}_e d^3x , \quad (51)$$

where ϕ_e is the scalar potential of the charges and $\mathbf{J}_e$ is the current density. In our case ϕ_e is the scalar potential of q and $\mathbf{J}_e$ is the current density of each monopole model given in equations (16), (29) and (41).

One subtle point about the expression for hidden mechanical momentum given in (51) is that it could be argued to be circular since for the field's linear momentum one can write

$$\begin{aligned} \mathbf{P}_{\text{EM}} &= \frac{1}{4\pi} \int (\mathbf{E} \times \mathbf{B}) d^3x = -\frac{1}{4\pi} \int (\nabla \phi_e \times \mathbf{B}) d^3x \\ &= -\frac{1}{4\pi} \int \nabla \times (\phi_e \mathbf{B}) d^3x + \frac{1}{4\pi} \int \phi_e \nabla \times \mathbf{B} d^3x \\ &= -\frac{1}{4\pi} \oint d\mathbf{a} \times (\phi_e \mathbf{B}) + \int \phi_e \mathbf{J}_e d^3x \end{aligned} \quad (52)$$

In arriving at the result in the last line we have used Stokes' theorem to obtain the first term and used $\nabla \times \mathbf{B} = 4\pi \mathbf{J}_e$ for the second term. Usually, the first surface term in (52) vanishes for localized sources and fields, by taking the surface to infinity. However, while the point electric source that produces ϕ_e does fall off with distance, the string part of $\mathbf{B}$ (see the second term in (13)) extends to infinity—taking that part to zero would be unjustified if we are to be rigorous. If one were to drop the surface term in (52), then one sees in general that $\mathbf{P}_{\text{hid}} = -\mathbf{P}_{\text{EM}}$ if (51) is used to calculate

the hidden momentum. One might argue for dropping the surface term in (52), despite the string part of $\mathbf{B}$ extending to infinity, since the string contribution is infinitesimally thin (which can be seen due to the $\delta(x)\delta(y)$ term). Alternatively, one could argue for dropping the surface term if the surface is taken to be spherical, since for a spherical surface $d\mathbf{a} \propto \hat{\mathbf{r}}$ and thus $d\mathbf{a} \times \mathbf{B} = 0$ along the z -axis of the spherical surface. However, in keeping with the formulation of Stokes's theorem, we will assume this term vanishes at the boundary as it is explicitly without bound in the positive or negative z -axis. For completeness, we present the explicit calculations confirming the cancellation of the electromagnetic field momentum of the three monopole models with the hidden mechanical momentum in Appendix C.

SUMMARY AND CONCLUSIONS

In this paper, we explored three magnetic monopole-like models: the Dirac string potential [6, 7], the Banderet potential [19], and the Wu-Yang fiber bundle approach [21, 22]. Each of these models has a non-Coulomb magnetic field. For the Dirac approach, the Dirac string is given by the second term in (13). For the Banderet approach there is a semi-infinite disk magnetic field given by the second term in (28). For the Wu-Yang approach there is the disk magnetic field in the xy -plane given by the second term in (30).

We have shown that these non-Coulomb magnetic fields plus the Coulomb electric field of q give rise to a non-zero field momentum despite the magnetic charge g and the electric charge, q , being at rest. By itself, this uncanceled field momentum – given in equations (43), (46) and (50) for the Dirac, Banderet, and Wu-Yang models, respectively – would violate the center-of-energy theorem [3, 26]. However, associated with each of these extra, non-Coulomb magnetic fields there was a current density $\mathbf{J}_e$, given in equations (16) (29) and (41) for the Dirac, Banderet and Wu-Yang cases, respectively. These current densities combined with the scalar potential, ϕ , of the electric charge q to give each of these systems a hidden mechanical momentum $\mathbf{P}_{\text{hid}}^{\text{mech}} = -\int \phi_e \mathbf{J} d^3x$. Calculating the hidden mechanical momentum for the three magnetic charge models – the results are given in (66) (74) and (78) for the Dirac, Banderet and Wu-Yang cases, respectively –

showed that the field momentum plus hidden mechanical momentum cancel each other out for the three magnetic charge models *i.e.* $\mathbf{P}_{\text{field}} + \mathbf{P}_{\text{mech}}^{\text{hid}} = 0$. This saves the center-of-energy theorem and reinforces the fact that the non-Coulomb magnetic fields of (13), (28) and (30) and their associated current densities of (16) (29) and (41) are needed for the internal consistency of these models. Previous works have argued for the physical reality of features such as the Dirac string based on their gravitational effects [30].

This analysis of magnetic monopole-like models serves as an introduction to the application of calculating hidden momentum of particle fields and shows a possible avenue of theoretical exploration for non-Abelian 't Hooft-Polyakov monopole models [31, 32] or recently investigated electroweak monopoles [33, 34]. As a preliminary to exploring the 't Hooft-Polyakov monopole that naturally arises in Yang-Mills theory (they are a solution to the equations, rather than added in by hand to Maxwellian Dynamics like the monopole-like models explored in this thesis), the Nielsen-Olesen flux tubes, which served as a structural precursor to that monopole model, will be explored in a forthcoming paper. Since this thesis has demonstrated the necessity for a consistent, physical model to uphold the center-of-energy theorem, we should expect, if these models are to correspond to real, physical object, that they should contain a hidden momentum which exactly cancels the field momentum that arises. We conclude that the intersection of theoretical physics, notably, particle physics, with the mathematics of differential geometry and topology presented in this

thesis highlights the deep interconnectedness between the fields of math and physics. We believe that further explorations of areas of physics, especially ones traditionally overlooked, through the lens of rigorous mathematical formulation, will continue to enrich both fields as well as our own academic pursuits.

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APPENDICES

APPENDIX A: MAGNETIC FLUX OF DELTA FUNCTION TERMS

In this appendix, we show that the magnetic flux of all three monopole models – Dirac, Banderet, and Wu-Yang – give an inward flux of $4\pi g$ to match the outward flux of $4\pi g$ due to the Coulomb part of the magnetic field.

First for the Coulomb part of the magnetic field (*i.e.* $\mathbf{B}_{\text{Coulomb}} = \frac{g\hat{\mathbf{r}}}{r^2}$) the flux through a sphere of radius R is given by

$$\oint \mathbf{B}_{\text{Coulomb}} \cdot d\mathbf{a} = \int gR^2 \frac{\hat{\mathbf{r}}}{R^2} \cdot \hat{\mathbf{r}} d\Omega = 4\pi g , \quad (53)$$

since the integral over the full solid angle is 4π .

For the Dirac version of the monopole, the string part of the magnetic field is $\pm 4\pi g \delta(x)\delta(y)\Theta(\mp z)\hat{\mathbf{z}}$ – see (13). The $\delta(x)\delta(y)$ term fixes the strings for the two potentials to run along the z -axis. For the string along the $-z$ axis the magnetic field is $+4\pi g \delta(x)\delta(y)\Theta(-z)\hat{\mathbf{z}}$ and $d\mathbf{a} = -dxdy\hat{\mathbf{z}}$ giving

$$\oint \mathbf{B}_{-}^{\text{string}} \cdot d\mathbf{a} = \int 4\pi g \delta(x)\delta(y)\hat{\mathbf{z}} \cdot (-\hat{\mathbf{z}}dxdy) = -4\pi g , \quad (54)$$

For the string along the $+z$ axis the magnetic field is $-4\pi g \delta(x)\delta(y)\Theta(+z)\hat{\mathbf{z}}$ and $d\mathbf{a} = dxdy\hat{\mathbf{z}}$ which gives

$$\oint \mathbf{B}_{+}^{\text{string}} \cdot d\mathbf{a} = \int -4\pi g \delta(x)\delta(y)\hat{\mathbf{z}} \cdot (\hat{\mathbf{z}}dxdy) = -4\pi g . \quad (55)$$

In both cases the string contribution gives a flux of $-4\pi g$ to balance the flux of $+4\pi g$ coming from the Coulomb part *i.e.*

$\Phi_{\text{Coulomb}} + \Phi_{\text{string}} = 4\pi g - 4\pi g = 0$. This is the flux through any closed surface containing the origin.

For the Banderet version of the monopole, the extra magnetic field given in (28) is $-2\pi g \Theta(x) \delta(y) x \frac{\hat{\mathbf{r}}}{r^2}$. We use a spherical surface of radius R to calculate the magnetic flux of this extra piece which gives

$$\begin{aligned} \oint \mathbf{B}^{\text{band}} \cdot d\mathbf{a} &= -2\pi g \int \Theta(x) \delta(y) x \frac{\hat{\mathbf{r}}}{R^2} \cdot \hat{\mathbf{r}} R^2 \sin \theta d\theta d\varphi \\ &= -2\pi g \int \Theta(R \sin \theta \cos \varphi) \delta(R \sin \theta \sin \varphi) R \sin^2 \theta \cos \varphi d\theta d\varphi. \end{aligned} \quad (56)$$

In the last step, we have converted everything to spherical polar coordinates.

Now we carry out the $d\varphi$ integration and use the delta function property

$\delta(f(x)) = \sum_i \frac{1}{|f'(x)|_{x=x_i}} \delta(x - x_i)$, where x_i are the zeros of $f(x)$ and $f'(x)$ is the derivative of $f(x)$. Now for the delta function in (56)

$f'(x) \rightarrow R \sin \theta \cos \varphi$ and the zeros occur at $\varphi = 0$ and $\varphi = \pi$. However,

because of the $\Theta(R \sin \theta \cos \varphi)$ term in (56) only $\varphi = 0$ is in the range of

integration since $\Theta(R \sin \theta \cos \varphi) = 0$ when $\varphi = \pi$. With all this the φ

integration is

$$\int_0^{2\pi} R \sin^2 \theta \cos \varphi \frac{1}{R \sin \theta \cos \varphi} \delta(\varphi - 0) d\varphi = \sin \theta \quad (57)$$

We now finish by evaluating the $d\theta$ integration in (56) which gives

$$\oint \mathbf{B}^{\text{band}} \cdot d\mathbf{a} = -2\pi g \int_0^\pi \sin \theta d\theta = -4\pi g . \quad (58)$$

Thus, the Coulomb magnetic flux and Banderet magnetic flux balance *i.e.*

$$\Phi_{\text{Coulomb}} + \Phi_{\text{band}} = 4\pi g - 4\pi g = 0.$$

For the Wu-Yang version of the monopole, the extra magnetic field given in (30) is $-\frac{2g\delta(z)}{\rho}\hat{\rho}$. We use a cylindrical surface of radius R and height $2H$ going from $z = -H$ to $z = H$. The magnetic flux due to the extra disk term is

$$\begin{aligned} & \oint \mathbf{B}^{\text{disk}} \cdot d\mathbf{a} \\ &= -2g \int \delta(z) \frac{\hat{\rho}}{R} \cdot \hat{\rho} R dz d\varphi = -2g \int_0^{2\pi} \int_{-H}^H \delta(z) dz d\varphi = -4\pi g , \quad (59) \end{aligned}$$

where the dz integration gives 1 by the property of the delta functions and the $d\varphi$ integration gives 2π . The end caps of the cylindrical surface contribute nothing, because $\mathbf{B} \perp d\mathbf{a}$. Thus, the Coulomb magnetic flux and disk magnetic flux balance *i.e.* $\Phi_{\text{Coulomb}} + \Phi_{\text{disk}} = 4\pi g - 4\pi g = 0$.

APPENDIX B: CURRENT DENSITY FOR BANDERET POTENTIAL

In this appendix we calculate the current density for the Banderet potential. The current density is given by $\mathbf{J}_e = \frac{\nabla \times \mathbf{B}}{4\pi}$. Using the magnetic field from (28) gives

$$\begin{aligned} \mathbf{J}_e &= -\frac{g}{2} \nabla \times \left(\Theta(x) \delta(y) x \frac{\hat{\mathbf{r}}}{r^2} \right) \\ &= \frac{g}{2} \frac{\hat{\mathbf{r}}}{r^2} \times (\delta(y) \delta(x) x \hat{\mathbf{x}} + \Theta(x) x \delta'(y) \hat{\mathbf{y}} + \Theta(x) \delta(y) \hat{\mathbf{x}}) \end{aligned} \quad (60)$$

In (60) we have used the product rule

$\nabla \times (f \mathbf{A}) = f \nabla \times \mathbf{A} - \mathbf{A} \times \nabla(f)$, with $\mathbf{A} = \frac{\hat{\mathbf{r}}}{r^2}$ and $f = \Theta(x) \delta(y) x$. Also $\nabla \times \frac{\hat{\mathbf{r}}}{r^2} = 0$ and the primes in (60) are derivatives with respect to the argument of the function. Now, the first term in (60) is zero since $x \delta(x)$ is zero. Next using $\hat{\mathbf{r}} = \frac{\mathbf{r}}{r}$ and $\mathbf{r} = x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}$ and $\mathbf{r} \times \hat{\mathbf{y}} = x \hat{\mathbf{z}} - z \hat{\mathbf{x}}$ and $\mathbf{r} \times \hat{\mathbf{x}} = -y \hat{\mathbf{z}} + z \hat{\mathbf{y}}$ the current density becomes

$$\begin{aligned} \mathbf{J}_e &= \frac{g \Theta(x)}{2r^3} (x^2 \delta'(y) \hat{\mathbf{z}} - x z \delta'(y) \hat{\mathbf{x}} - y \delta(y) \hat{\mathbf{z}} + z \delta(y) \hat{\mathbf{y}}) \\ &= \frac{g \Theta(x)}{2r^3} (x^2 \delta'(y) \hat{\mathbf{z}} - x z \delta'(y) \hat{\mathbf{x}} + z \delta(y) \hat{\mathbf{y}}) , \end{aligned} \quad (61)$$

where in the last term we have used $y \delta(y) = 0$.

APPENDIX C: HIDDEN MECHANICAL MOMENTUM FOR THREE MAGNETIC CHARGE MODELS

This appendix presents explicit calculations for the hidden mechanical momentum for the Dirac, Banderet, and Wu-Yang models for magnetic charge.

Hidden Mechanical Momentum of Dirac String

To calculate the hidden momentum for the Dirac string case we need the scalar potential, ϕ , for the electric charge and the current density for the Dirac string. The current density was calculated in section IIA and the result is given in equation (16) as $\mathbf{J}_e = \pm g \Theta(\mp z) [\delta(x) \delta'(y) \hat{\mathbf{x}} - \delta(y) \delta'(x) \hat{\mathbf{y}}]$. Due to the cylindrical symmetry of the Dirac string potential we can place, without loss of generality, the electric charge anywhere in the xy -plane on a ring of radius $\rho_0^2 = x_0^2 + y_0^2$. Thus we will choose the electric charge, q , to be on the x -axis at $x_0 = \rho_0$ and along the z -direction at $z = z_0$. This leaves $y_0 = 0$. With these choices the scalar potential becomes

$$\phi_e = \frac{q}{\sqrt{(x - x_0)^2 + y^2 + (z - z_0)^2}} . \quad (62)$$

With this setup the hidden mechanical momentum for the Dirac string case, with the string along the negative z -axis, becomes

$$\mathbf{P}_{\text{hid}}^{\text{String}} = -gq \int_{-\infty}^0 dz \int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dx \frac{[\delta(x)\delta'(y)\hat{\mathbf{x}} - \delta(y)\delta'(x)\hat{\mathbf{y}}]}{\sqrt{(x-x_0)^2 + y^2 + (z-z_0)^2}} . \quad (63)$$

The $\Theta(-z)$ function in the current density has been taken into account via the limits in the dz integration. The case with the string along the positive z -axis works out identically except for some changes in signs. To handle the $\delta'(y)$ and $\delta'(x)$ terms we use integration by parts. For the $\delta'(y)$ term the dy integration by parts gives

$$\left. \frac{\delta(y)}{\sqrt{(x-x_0)^2 + y^2 + z^2}} \right|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{y\delta(y)}{((x-x_0)^2 + y^2 + (z-z_0)^2)^{3/2}} dy = 0 . \quad (64)$$

Both terms in (64) are zero since $\delta(\pm\infty) = 0$, and thus the $\hat{\mathbf{x}}$ term in (63) is zero. For the $\delta'(x)$ term the dx integration by parts gives

$$\begin{aligned} & \left. \frac{\delta(x)}{\sqrt{(x-x_0)^2 + y^2 + (z-z_0)^2}} \right|_{-\infty}^{\infty} \\ & + \int_{-\infty}^{\infty} \frac{(x-x_0)\delta(x)}{((x-x_0)^2 + y^2 + (z-z_0)^2)^{3/2}} dx \\ & = -\frac{x_0}{(x_0^2 + y^2 + (z-z_0)^2)^{3/2}} . \end{aligned} \quad (65)$$

The first term in (65) is zero since $\delta(\pm\infty) = 0$, but the second term now gives a non-zero part. Having carried out the dx -integration for the second term in (63), we can immediately carry out the dy -integration with the help

of $\delta(y)$, leaving only the dz integration

$$\begin{aligned}
\mathbf{P}_{\text{hid}}^{\text{String}} &= -gqx_0\hat{\mathbf{y}} \int_{-\infty}^0 \frac{dz}{(x_0^2 + (z - z_0)^2)^{3/2}} \\
&= -gqx_0\hat{\mathbf{y}} \left(\frac{1}{x_0^2} - \frac{z_0}{x_0^2 \sqrt{x_0^2 + z_0^2}} \right) \\
&= -gq \frac{(r_0 - z_0)}{r_0 x_0} \hat{\mathbf{y}} \rightarrow -gq \frac{(r_0 - z_0)}{r_0 \rho_0} \hat{\mathbf{y}} .
\end{aligned} \tag{66}$$

In (66) we have taken into account that $x_0 \rightarrow \rho_0$ since we choose $y_0 = 0$, and also $r_0 = \sqrt{x_0^2 + z_0^2} \rightarrow \sqrt{\rho_0^2 + z_0^2}$. For the case where the string is along the positive z -axis the results are similar to (66) but with some changes in sign giving $\mathbf{P}_{\text{hid}}^{\text{String}} = -gq \frac{(r_0 + z_0)}{r_0 \rho_0} \hat{\mathbf{y}}$

In the expression for the field momentum from (43) if we let $x_0 \rightarrow \rho_0$ and $y_0 \rightarrow 0$ we get $\mathbf{P}_{\text{EM}}^{\text{String}} = gq \frac{(r_0 \mp z_0)}{r_0 \rho_0} \hat{\mathbf{y}}$. Combining this field momentum with the hidden string momentum from (66) gives $\mathbf{P}_{\text{EM}}^{\text{String}} + \mathbf{P}_{\text{hid}}^{\text{String}} = 0$ thus preserving the center of energy theorem.

Hidden Mechanical Momentum of Banderet Potential

For the field momentum calculation for the Banderet potential given in section IIIB we placed the electric charge, q , at $\mathbf{r}_0 = y_0 \hat{\mathbf{y}}$ in order to get a closed-form analytical result. This gives a scalar potential of

$$\phi_e = \frac{q}{\sqrt{x^2 + (y - y_0)^2 + z^2}} . \tag{67}$$

The current density for this case was calculated in equation (29) and the result is given by $\mathbf{J} = \frac{g\Theta(x)}{2r^3} (x^2\delta'(y)\hat{\mathbf{z}} - xz\delta'(y)\hat{\mathbf{x}} + z\delta(y)\hat{\mathbf{y}})$. The hidden momentum for the Banderet model is thus

$$\begin{aligned} & \mathbf{P}_{\text{hid}}^{\text{Banderet}} \\ = & -\frac{qg}{2} \int \frac{\Theta(x)}{r^3 \sqrt{x^2 + (y - y_0)^2 + z^2}} \\ * & (x^2\delta'(y)\hat{\mathbf{z}} - xz\delta'(y)\hat{\mathbf{x}} + z\delta(y)\hat{\mathbf{y}}) dx dy dz \end{aligned} \quad (68)$$

The dz integration for $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ components of $\mathbf{P}_{\text{hid}}^{\text{Banderet}}$ are of the form

$$\int_{-\infty}^{\infty} \frac{z}{(x^2 + (y - y_0)^2 + z^2)^{1/2} (x^2 + y^2 + z^2)^{3/2}} dz = 0 \quad (69)$$

Since the integrand is odd and the integration range is over all of z , symmetry gives zero for the integration. Thus, $\mathbf{P}_{\text{hid}}^{\text{Banderet}}$ has no $\hat{\mathbf{x}}$ or $\hat{\mathbf{y}}$ components.

Moving on to the $\hat{\mathbf{z}}$ component of $\mathbf{J}_e$ and $\mathbf{P}_{\text{hid}}^{\text{Banderet}}$, we evaluate the dy integration via integration by parts

$$\begin{aligned}
& \int_{-\infty}^{\infty} \frac{\delta'(y) dy}{(x^2 + (y - y_0)^2 + z^2)^{1/2} (x^2 + y^2 + z^2)^{3/2}} \\
&= \left. \frac{\delta(y)}{(x^2 + (y - y_0)^2 + z^2)^{1/2} (x^2 + y^2 + z^2)^{3/2}} \right|_{-\infty}^{\infty} \\
&- \int_{-\infty}^{\infty} \frac{-3y\delta(y) dy}{(x^2 + (y - y_0)^2 + z^2)^{1/2} (x^2 + y^2 + z^2)^{5/2}} \\
&- \int_{-\infty}^{\infty} \frac{\delta(y)(y - y_0) dy}{(x^2 + (y - y_0)^2 + z^2)^{3/2} (x^2 + y^2 + z^2)^{3/2}} \\
&= -\frac{y_0}{(x^2 + y_0^2 + z^2)^{3/2} (x^2 + z^2)^{3/2}} . \tag{70}
\end{aligned}$$

The first two terms on the right-hand side of (70) are zero by the property $\delta(\pm\infty) = 0$.

Using the result of (70) the hidden momentum for the Banderet potential is

$$\mathbf{P}_{\text{hid}}^{\text{Banderet}} = \frac{gqy_0\hat{\mathbf{z}}}{2} \int_0^\infty x^2 dx \int_{-\infty}^\infty \frac{dz}{(x^2 + z^2)^{3/2} (x^2 + y_0^2 + z^2)^{3/2}} . \tag{71}$$

In (71) we have taken into account the $\Theta(x)$ term in $\mathbf{J}$ in the range of the dx integration. The dx integration can be done via *Mathematica* and yields

$$\begin{aligned}
& \int_0^\infty \frac{x^2}{(x^2 + z^2)^{3/2} (x^2 + y_0^2 + z^2)^{3/2}} dx \\
&= \frac{-2z^2 E(-y_0^2/z^2) + (y_0^2 + 2z^2) K(-y_0^2/z^2)}{y_0^4 |z|} , \tag{72}
\end{aligned}$$

where $K(x)$ and $E(x)$ are complete elliptic integrals of the first and second kind. Using *Mathematica* to carry out the dz integration of (72) yields

$$\int_{-\infty}^{\infty} \frac{-2z^2 E(-y_0^2/z^2) + (y_0^2 + 2z^2)K(-y_0^2/z^2)}{y_0^4|z|} dz = \frac{\pi}{2y_0^2} . \quad (73)$$

Inserting the result of (72) and (73) into (71) gives a hidden momentum for the Banderet model of the form

$$\mathbf{P}_{\text{hid}}^{\text{Banderet}} = \frac{gq\pi}{4y_0} \hat{\mathbf{z}} . \quad (74)$$

Combining this hidden momentum of the Banderet model in (74) with the Banderet field momentum result of (46) we find $\mathbf{P}_{\text{EM}}^{\text{Banderet}} + \mathbf{P}_{\text{hid}}^{\text{Banderet}} = 0$ and thus the center-of-energy theorem is preserved.

Hidden Mechanical Momentum of Wu-Yang Fiber Bundle

For the Wu-Yang fiber bundle approach, the current density is given in (41) and has the form $\mathbf{J}_e = -\frac{q}{2\pi\rho}\delta'(z)\hat{\phi}$. As in the case of the Dirac string, the cylindrical symmetry of the disk singularity of the Wu-Yang case allows one (without loss of generality) to place the electric charge, q , anywhere in the xy -plane at a fixed radius of ρ_0 . We again choose q to be on the x -axis at $x = \rho_0$ and at a position of $z = z_0$ along the z -direction. The magnetic potential for q is thus the same as for the Dirac string case in (62) namely $\phi_e = \frac{q}{\sqrt{(x-\rho_0)^2 + y^2 + (z-z_0)^2}}$. Combining these results for the current density and scalar potential, and writing things in cylindrical coordinates the hidden

momentum for the disk in the Wu-Yang set-up gives

$$\mathbf{P}_{\text{hid}}^{\text{Disk}} = \frac{gq}{2\pi} \int \frac{\delta'(z)(-\sin \varphi \hat{\mathbf{x}} + \cos \varphi \hat{\mathbf{y}})}{\rho \sqrt{\rho^2 + \rho_0^2 + (z - z_0)^2 - 2\rho_0 \rho \cos \varphi}} \rho d\rho d\varphi dz . \quad (75)$$

For the $\hat{\mathbf{x}}$ component in (75) the $d\varphi$ integration is of the form

$$\int_0^{2\pi} \frac{\sin \varphi}{\sqrt{A - B \cos \varphi}} d\varphi \text{ with } A \equiv \rho^2 + \rho_0^2 + (z - z_0)^2 \text{ and } B \equiv 2\rho_0 \rho. \text{ Using}$$

Mathematica this integration is zero, thus eliminating the $\hat{\mathbf{x}}$ component in

$$\mathbf{P}_{\text{hid}}^{\text{Disk}} .$$

Moving on to the $\hat{\mathbf{y}}$ component in (75) we first carry out the dz integration using integration by parts yielding

$$\begin{aligned} & \frac{gq \cos \varphi}{2\pi \rho} \hat{\mathbf{y}} \int_{-\infty}^{\infty} \frac{\delta'(z)}{\sqrt{K + (z - z_0)^2}} dz \\ &= \frac{gq \cos \varphi}{2\pi \rho} \hat{\mathbf{y}} \left[\frac{\delta(z)}{\sqrt{K + (z - z_0)^2}} \Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{\delta(z)(z - z_0)}{(K + (z - z_0)^2)^{3/2}} dz \right] \\ &= \frac{gq \cos \varphi}{2\pi \rho} \left(\frac{-z_0}{(K + z_0^2)^{3/2}} \right) \hat{\mathbf{y}} . \end{aligned} \quad (76)$$

In (76) we have defined $K \equiv \rho^2 + \rho_0^2 - 2\rho_0 \rho \cos \varphi$. This first surface term in the middle line of (76) is zero by the property $\delta(\pm\infty) = 0$. We now carry out the $d\rho$ integration via *Mathematica* which yields

$$\begin{aligned} & - \frac{gq z_0 \cos \varphi}{2\pi} \hat{\mathbf{y}} \int_0^{\infty} \frac{1}{\rho(\rho^2 + \rho_0^2 + z_0^2 - 2\rho_0 \rho \cos \varphi)^{3/2}} \rho d\rho \\ &= - \frac{gq z_0 \cos \varphi}{2\pi} \hat{\mathbf{y}} \left(\frac{2(1 + \rho_0 \cos \varphi / r_0)}{\rho_0^2 + 2z_0^2 - \rho_0^2 \cos 2\varphi} \right) , \end{aligned} \quad (77)$$

Finally we carry out the $d\varphi$ integration using *Mathematica* yielding

$$\begin{aligned}\mathbf{P}_{\text{hid}}^{\text{Disk}} &= -\frac{gqz_0}{2\pi}\hat{\mathbf{y}}\int_0^{2\pi}\left(\frac{2\cos\varphi(1+\rho_0\cos\varphi/r_0)}{\rho_0^2+2z_0^2-\rho_0^2\cos 2\varphi}\right)d\varphi \\ &= -\frac{gqz_0}{2\pi}\hat{\mathbf{y}}\left(\frac{-2\pi}{\rho_0r_0}\right)=\frac{qgz_0}{\rho_0r_0}\hat{\mathbf{y}}\ .\end{aligned}\tag{78}$$

Adding the result in (78) for the hidden mechanical momentum to the field momentum in (50) one finds $\mathbf{P}_{\text{mech}}^{\text{hid}} + \mathbf{P}_{\text{EM}}^{\text{Disk}} = 0$ and thus the center of energy theorem is satisfied.

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